

Question number	Answer	Additional guidance	Mark
8(a)(i)	- total number of deaths in one year calculated (correctly rounded) (1) $\{6 / 5.8 / 5.77\} \times 10^7$ (1)	57.74193548387 million / $5.774193548387 \times 10^7$ Bald correct answer = 2 marks Bald answer of total number of deaths in one year correctly rounded = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
8(b)(i)	An explanation that includes the following points: • because the antioxidants will reduce free radicals (1) • therefore {cell damage / damage to lining of blood vessels / oxidative stress} will be {reduced / prevented} (1) • (less cell damage / antioxidants) {reduce/ prevent} {plaque / atheroma} formation (1)	ACCEPT neutralise / donate electrons to / break down / stabilise ACCEPT reduces cholesterol build up	(3)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	• not harmful / food source / already in diet (1)		(1)

Question number	Answer	Additional guidance	Mark
*8(b)(iii)	Indicative content: For repeatable data: • a large group of people (500 minimum) should be selected (D) • study could be repeated (D) • uses same methods for all {groups / trials} (D) For valid data: • people should be selected who have no underlying health risks (D) • other CVD risk factors should be controlled (D) • use of medication should be controlled (D) • as they should all have similar risk profiles (E) • e.g. shouldn't smoke as this would increase blood pressure (E) • diet should be controlled (D) • so that they are consuming the same number of free radicals (E) • so that they are consuming the same number of antioxidants (E) • so that they are consuming the same {cholesterol / HDL / LDL} • the ginger should be administered in a palatable and regular way (D) • so that everyone is getting the same mass of ginger (E) • one group of people should be given a placebo / known antioxidant (D) • to {act as a comparison to / to determine} the effect of the ginger (E) • people should be monitored for the development of CVD over a number (5 minimum) of years (D) • because CVD takes a period of time to develop (E) • use of statistical analysis / t-test / other relevant named stats test (D) • to check repeatability (E)	Level 1: Description of study design 1 mark = one appropriate description 2 marks = three appropriate descriptions Level 2: Description of study design (that will work) with some explanations 3 marks = three appropriate descriptions plus an indication that the study needs to be long term OR one description explained or linked correctly to either validity or repeatability 4 marks = two descriptions explained or linked correctly to both validity and repeatability Level 3: Descriptions with detailed explanations 5 marks = three descriptions explained 6 marks = at least three descriptions explained, including an indication that the study needs to be long-term with a reason	(6)